ProtCOVER: Family-Aware Conformal Prediction for Variant Effect Predictors on Unseen Protein Families

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Abstract. Protein language models and related variant effect predictors (VEPs) are increasingly used to classify missense variants, yet their reliability is typically summarized by benchmark-wide averages. In practice, predictors are applied to proteins that are evolutionarily distant from those used for calibration, which constitutes a family-level distribution shift. Standard split conformal prediction assumes exchangeable variants and offers no guarantee when entire protein families are unseen, while existing VEP calibration schemes condition on residue, gene or domain attributes rather than on homology support. We present ProtCOVER, a conformal procedure that treats homology families, rather than individual variants, as the exchangeable unit. ProtCOVER (i) constructs label-free homology clusters and calibrates at the family level, yielding marginal coverage for a new family under family exchangeability; (ii) reweights calibration families using estimated family-level density ratios from label-free descriptors (sequence similarity, alignment depth and embedding distance) to adapt to descriptor shift; and (iii) computes the effective calibration support for each query protein and returns a conservative fallback set, or abstains, when support is insufficient. We state the assumptions under which each guarantee holds and separately measure empirical robustness where they do not. We evaluate ProtCOVER on the ProteinGym clinical substitution benchmark using precomputed zero-shot scores from five predictors under leave-family-out splits, against split, Mondrian, group-weighted and hierarchical conformal baselines. Class-conditional split conformal, calibrated over variants, covers only 0.749–0.790 of pathogenic variants per family for four of the five predictors at a 0.90 target, even without shift. Under descriptor shift, family-level calibration alone falls to 0.798, while ProtCOVER's family-level weighting keeps family-weighted pathogenic and benign coverage at or above 0.886 for every predictor across four shifted settings, at the cost of larger prediction sets. Its gains come from shifts in alignment depth and sequence similarity; on orphan families and on held-out regions of embedding space it brings no gain over the group-weighted and family-level baselines. The support gate preserves these guarantees, but on this benchmark support-based abstention did not reduce the error of retained predictions relative to random abstention. Code is publicly available.

Keywords: conformal prediction · variant effect prediction · protein language models · uncertainty quantification · distribution shift · selective prediction · ProteinGym · protein families

1 Introduction

Zero-shot variant effect predictors (VEPs), from protein language models such as ESM-1b [30,6] to alignment-based models such as EVE and GEMME [12,16] and hybrids such as TranceptEVE and PoET [25,34], now rank missense variants with high accuracy on clinical benchmarks [24]. Clinical use, however, needs more than a ranking: variant interpretation under the guidelines of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) [29] needs evidence whose error rate is known, and the ClinGen calibration of computational evidence [27] has shown that a single genome-wide calibration can hide large differences between genes [32].

Conformal prediction [36,1] offers distribution-free error control: a prediction set over $\{\text{benign}, \text{pathogenic}\}$ that contains the true label with probability at least $1 - \alpha$. Its guarantee rests on exchangeability between calibration and test data. For VEPs this assumption fails in two ways. First, variants are grouped into proteins and homologous protein families: a predictor's accuracy differs from family to family, and families contribute very different numbers of labeled variants (from 2 to 741 per protein in our data). Pooling variants lets the largest families set the error rate, so the "90%" is a statement about variants from well-studied genes, not about a new protein. Second, the proteins a predictor is deployed on are often evolutionarily distant from, or

have shallower alignments than, the proteins in the calibration set: a family-level distribution shift that is visible without labels.

We make three contributions.

- We formalize *family-weighted* coverage, the probability that a random variant of a new protein family is covered, and its class-conditional version for pathogenic variants. On ProteinGym clinical substitutions, class-conditional (Mondrian) split conformal, the standard remedy for class imbalance, meets its own variant-level target yet covers only 0.749–0.790 of pathogenic variants per family for four of five predictors at a 0.90 target, with no distribution shift at all.
- We present ProtCOVER, which (i) calibrates at the family level, (ii) reweights calibration families by a cross-fitted density ratio of label-free family descriptors, and (iii) gates queries with insufficient calibration support. We prove finite-sample family-weighted coverage under family exchangeability (Proposition 1, a restatement of hierarchical conformal prediction [18]) and under family-level covariate shift (Theorem 1), bound the loss from estimated weights (Corollary 1), and show that the gate preserves all of these (Lemma 1).
- We evaluate five predictors under exchangeable, synthetic-shift, orphan-family, and held-out-region splits with family-level bootstrap intervals. Under shift, ProtCOVER keeps both class-conditional coverages at or above 0.886 for every predictor and setting, against 0.798 for family-level calibration alone and 0.873 for group-weighted conformal prediction. The gain comes from shifts that act on the descriptors it models; on orphan families and held-out regions of embedding space it brings no gain over these baselines, and it costs larger sets. We also report a negative result: abstaining on poorly supported families did not reduce the error of the remaining predictions relative to random abstention.

2 Related Work

Conformal prediction. Split conformal prediction [36,19] gives marginal coverage under exchangeability; Mondrian or class-conditional versions calibrate separately within categories [35,9]. Weighted conformal prediction handles covariate shift with a known or estimated likelihood ratio [33], with coverage loss bounded by the ratio's estimation error [20]; related reweightings address label shift [28] and general departures from exchangeability [2]. Group-weighted conformal prediction (GWCP) [3] sharpens these bounds when the shift acts through a finite set of known groups; we use it with descriptor bins as a baseline. In protein science, conformal methods have handled the feedback covariate shift of sequence design [11] and given error guarantees for mining protein databases for function [5]. Localized conformal prediction [15] reweights calibration points by proximity to the query, and robust and multi-source methods protect against families of target distributions [7,22]. For grouped data, Dunn et al. [10] proposed pooling and subsampling methods for a new group, and Lee et al. [18] proved finite-sample guarantees for the pooled construction (HCP); later work uses a few observations from the test group [23]. Macro-coverage [4] averages coverage over classes rather than over samples, in the same spirit as our family weighting. ProtCOVER combines family-level pooling with family-level density-ratio weights, which none of these works does, and applies it to protein families defined by homology.

Calibration of variant effect predictors. ClinGen calibrated computational scores to evidence strengths for the computational evidence codes PP3 (supporting pathogenicity) and BP4 (supporting benignity) [27]; subsequent work found that genome-wide calibration masks heterogeneous performance across genes [32] and proposed gene- and domain-aware calibration [8], and RaCoon calibrates ESM-1b at the residue level with conditional score distributions [26]. These methods improve calibration of point probabilities within subgroups seen in training; they do not give distribution-free guarantees, and they condition on gene, domain or residue attributes rather than on homology support relative to the calibration data. Kermut [14] provides calibrated uncertainty for supervised variant effect regression on deep mutational scans. ProteinGym [24] supplies the benchmark and precomputed scores we use, including those of PLM-based and alignment-based predictors [30,12,16,25,34] and of supervised clinical predictors, which we exclude because they are trained on ClinVar labels [17]. Selective classification [13] motivates our risk-coverage evaluation of abstention.

3 Problem Setup

Families and scores. A protein *family* is a cluster of homologous wild-type sequences obtained without labels (Section 6.1). Family k is the random object $G_k = (D_k, N_k, (X_{k,i}, Y_{k,i})_{i=1}^{N_k})$, where $X_{k,i}$ describes a missense variant, $Y_{k,i} \in \{0,1\}$ is its label (1 = pathogenic), and $D_k \in \mathcal{D}$ is a *label-free descriptor* of the family: any function of its sequences, alignments, predictor scores, and of the fixed set of training families, but not of labels. A predictor score is turned into $\hat{p}(y \mid x)$ by a base model fitted on separate training families, which we treat as fixed. The nonconformity score is $s(x,y) = 1 - \hat{p}(y \mid x)$ and $S_{k,i} = s(X_{k,i}, Y_{k,i})$. We observe calibration families $G_1, \ldots, G_K$ with labels and a test family G_{K+1} whose descriptor D_{K+1} and variants $X_{K+1,i}$ are observed but whose labels are not.

Coverage targets. For a set-valued predictor $C : \mathcal{X} \to 2^{\{0,1\}}$, the per-family coverage of the test family is $\gamma_{K+1}(C) = N_{K+1}^{-1} \sum_i \mathbf{1}\{Y_{K+1,i} \in C(X_{K+1,i})\}$. Our primary target is *family-weighted coverage*

$$\mathrm{Cov}(C) = \mathbb{E}\big[\gamma_{K+1}(C)\big], \tag{1}$$

the probability that a uniformly random variant of a new family is covered; every family counts once, whatever its number of variants. Its class-conditional version for label y is $\mathrm{Cov}_y(C) = \mathbb{E}\big[N_{K+1,y}^{-1} \sum_{i:Y_{K+1,i}=y} \mathbf{1}\{y \in C(X_{K+1,i})\} \,\big|\, N_{K+1,y} > 0\big]$, where $N_{k,y}$ counts label-y variants of family k. For clinical use, Cov_1 (pathogenic coverage) is the quantity of interest: $1 - \mathrm{Cov}_1$ is the family-weighted *pathogenic miss rate*, the rate at which a pathogenic variant's set excludes the pathogenic label (it is called benign or, rarely, receives an empty set). Variant-weighted coverage, which pools all test variants, lets families with many variants dominate; we report it but do not target it. Coverage for each individual family cannot be guaranteed without assumptions on how families differ [35], so the average over new families is the natural distribution-free target.

Assumptions. We use two alternative assumptions; each result states which one it needs.

A1 (*Family exchangeability.*) $(G_1, \ldots, G_{K+1})$ is exchangeable.

A2 (*Family-level covariate shift.*) $G_1, \ldots, G_K \overset{\text{iid}}{\sim} P$ and, independently, the test family G_{K+1} and any further unlabeled target families are i.i.d. from Q, where $Q \ll P$ and $\frac{dQ}{dP}(G) = w(D)$ for a measurable $w : \mathcal{D} \to [0, \infty)$. Equivalently, the conditional law of a family's variants and labels given its descriptor is the same under P and Q; only the distribution of descriptors shifts.

A1 holds when calibration and test families are drawn by a random split of a pool of families, regardless of dependence between families. It does not require exchangeability of variants within a family, because Cov averages over all variants of the test family (this is weaker than the hierarchical exchangeability of [18], which targets one variant of the new group). A2 formalizes deployment on families that differ from the calibration families in label-free ways, such as being evolutionarily distant or having shallow alignments.

4 ProtCOVER

(i) Family-level calibration. Let $\hat{F}_k = N_k^{-1} \sum_i \delta_{S_{k,i}}$ be the empirical distribution of calibration scores of family k. Given nonnegative family weights $\omega_1, \ldots, \omega_{K+1}$, set $p_k = \omega_k / \sum_{j=1}^{K+1} \omega_j$ and

$$\hat{\mu} = \sum_{k=1}^{K} p_k \hat{F}_k + p_{K+1}\, \delta_{+\infty}, \qquad \hat{q} = \mathrm{Quant}_{1-\alpha}(\hat{\mu}), \qquad C(x) = \{y \in \{0,1\} : s(x,y) \le \hat{q}\}, \tag{2}$$

where $\mathrm{Quant}_\tau(\mu) = \inf\{u : \mu((-\infty, u]) \ge \tau\}$. With $\omega \equiv 1$ each calibration family has total weight $1/(K+1)$ spread evenly over its variants, and the test family's mass sits at $+\infty$; this is the hierarchical conformal prediction (HCP) threshold of [18], which equals the CDF-pooling method of [10] up to the level adjustment. Variant-level split conformal instead gives every calibration variant weight 1, so families with hundreds of variants dominate the quantile.

(ii) Family-level weighting. To adapt to descriptor shift, ProtCOVER sets $\omega_k = \hat{w}(D_k)$ for an estimate $\hat{w}$ of the density ratio $w = dQ/dP$ on descriptors, and $\omega_{K+1} = \hat{w}(D_{K+1})$; the test family's mass is computable because D_{K+1} is label-free. The estimate comes from a logistic classifier separating calibration families (class 0) from target families (class 1) on standardized descriptors: $\hat{w}(d) = \frac{n_0}{n_1} \exp\{\hat{f}(d)\}$ with $\hat{f}$ the fitted log-odds and n_0, n_1 the numbers of calibration and target families in the fit, L_2 regularization chosen by five-fold cross-validated log loss on this membership task (no variant labels), weights rescaled to mean one over the calibration families used to fit $\hat{w}$, and clipped to $[1/c, c]$ ($c = 20$). We use descriptors for sequence similarity (maximum identity of the family to any training protein, over significant MMseqs2 [31] hits), alignment depth ($\log N_{\text{eff}}$ and N_{eff} per position, where N_{eff} is the effective number of sequences in the alignment), and embedding distance (cosine distance of mean-pooled ESM2 [21] embeddings to the nearest training protein), together with log length and log number of scored variants.

Cross-fitting. The guarantee for estimated weights (Corollary 1) needs $\hat{w}$ to be independent of the families it weights. We therefore split the calibration families once, at random, into a third used only to fit $\hat{w}$ and two thirds used in (2), and partition the target families into five folds; the ratio applied to fold j (and to the calibration families, for the queries of fold j) is fitted against the target families outside fold j.

(iii) Support gate. The gate acts on the family of a query protein, so all variants of that family share one decision. For a query family t with standardized support descriptors $\tilde{d}_t$, the localized calibration weights $\kappa_{tk} = \exp(-\|\tilde{d}_k - \tilde{d}_t\|^2/2b^2)$ give the effective calibration support (an effective sample size)

$$\mathrm{ESS}_t = \Big(\sum_k \kappa_{tk}\Big)^2 \Big/ \sum_k \kappa_{tk}^2, \tag{3}$$

the effective number of calibration families that resemble the query; the test-point mass $\pi_t = p_{K+1}$ measures how atypical the query is under the weights of (ii). The gate returns the conservative fallback set $\{0, 1\}$, that is, it abstains, when $\mathrm{ESS}_t < m$ or $\pi_t > \pi_{\max}$. The bandwidth b (a fraction of the median pairwise descriptor distance), m, and $\pi_{\max}$ are tuned only on an inner split of the training families (Section 6.2).

(iv) Class-conditional versions. Each component has a Mondrian version: for each candidate label y, the threshold $\hat{q}_y$ is computed from (2) using only label-y calibration variants ($\hat{F}_k^y$ over the $N_{k,y}$ label-y variants, families with $N_{k,y} = 0$ dropped), and $y \in C(x)$ iff $s(x, y) \le \hat{q}_y$. These target Cov_0 and Cov_1 separately.

5 Guarantees

All results are finite-sample and distribution-free given the stated assumption; proofs are in Appendix A. They hold for any fixed score function, so they do not depend on the quality of the base model or of the predictor.

Proposition 1 (Family-level calibration). *Under A1, the set* (2) *with* $\omega \equiv 1$ *satisfies* $\mathrm{Cov}(C) \ge 1 - \alpha$.

Proposition 1 is Theorem 1 of [18] with the test point replaced by a uniformly random variant of the test family; we include the short proof because it removes the need for within-family exchangeability.

Theorem 1 (Family-level weighted calibration). *Under A2, the set* (2) *with* $\omega_k = w(D_k)$, $k = 1, \ldots, K+1$, *satisfies* $\mathrm{Cov}(C) \ge 1 - \alpha$, *where the expectation is over* $G_1, \ldots, G_K \sim P$ *and* $G_{K+1} \sim Q$.

Corollary 1 (Estimated weights). *Under A2, let* $\hat{w}$ *be independent of* $(G_1, \ldots, G_{K+1})$ *with* $0 < \mathbb{E}_P \hat{w}(D) < \infty$, *and let* $\hat{Q}$ *be defined by* $d\hat{Q}/dP = \hat{w}(D)/\mathbb{E}_P \hat{w}(D)$. *With* $\omega_k = \hat{w}(D_k)$, *conditionally on* $\hat{w}$,

$$\mathbb{E}\big[\gamma_{K+1}(C) \,\big|\, \hat{w}\big] \;\ge\; 1 - \alpha - d_{\mathrm{TV}}(Q, \hat{Q}) \;=\; 1 - \alpha - \tfrac{1}{2}\,\mathbb{E}_P\Big|w(D) - \frac{\hat{w}(D)}{\mathbb{E}_P \hat{w}(D)}\Big|,$$

and hence $\mathrm{Cov}(C) \ge 1 - \alpha - \mathbb{E}\, d_{\mathrm{TV}}(Q, \hat{Q})$.

Theorem 1 lifts the weighted-exchangeability argument of [33] from variants to families, and Corollary 1 lifts [20] (their Theorem 3); what is new is that weights and test mass act on whole families, so the guarantees concern family-weighted coverage. Under cross-fitting, for queries in fold j the fitted ratio is independent of the calibration families used in (2) and of the query family, so the corollary applies fold by fold. Rescaling and clipping use only the fitting families, so the clipped ratio is again a fixed function of D given the fitting data, and the corollary applies to it directly.

Proposition 2 (Class-conditional versions). *For each $y \in \{0,1\}$ with $\mathbb{P}(N_{K+1,y} > 0) > 0$, the Mondrian versions of Proposition 1 and Theorem 1 hold with* Cov *replaced by* Cov_y*, and that of Corollary 1 holds with* Cov_y *and with $d_{\mathrm{TV}}(Q_y, \hat{Q}_y)$ in place of $d_{\mathrm{TV}}(Q, \hat{Q})$, where Q_y, $\hat{Q}_y$ are Q, $\hat{Q}$ conditioned on $N_{K+1,y} > 0$.*

Lemma 1 (Enlargement). *If $C'(x) \supseteq C(x)$ for all x, almost surely, where C' may depend on all observed data, then $\mathrm{Cov}(C') \geq \mathrm{Cov}(C)$ and $\mathrm{Cov}_y(C') \geq \mathrm{Cov}_y(C)$.*

Because the gate (iii) only replaces a set by $\{0,1\}$, Lemma 1 shows that it preserves every guarantee above, whatever its parameters.

What is and is not guaranteed. The guarantees are marginal over families. They do not imply coverage for a particular family, nor coverage on the queries the gate retains (a selected subset). Corollary 1 bounds the loss from estimated weights by a quantity that is not observable. In S1, families are split at random, so A1 and Proposition 1 hold exactly. Our S2 splits sample families without replacement from a finite pool, and S3 and S4 have no explicit shift model, so A2 holds only approximately, if at all, there. Everything we report about per-family dispersion, retained queries, S2, S3, and S4 is therefore empirical; Section 6.4 checks the guarantees themselves on synthetic data where A1 or A2 hold exactly.

6 Experiments

6.1 Data, predictors, and families

We use the ProteinGym v1.3 clinical substitution benchmark [24]: 62,727 ClinVar missense variants in 2,525 human proteins, 51.0% pathogenic, with a median of 12 labeled variants per protein (maximum 741). We take ProteinGym's precomputed zero-shot scores for five unsupervised predictors: ESM-1b (protein language model), EVE and GEMME (alignment-based), TranceptEVE (hybrid), and PoET (retrieval-augmented language model). We orient every score so that higher means more pathogenic, using ProteinGym's directionality settings; our per-protein mean AUCs agree with ProteinGym's published values (0.892–0.920) to three decimals. Supervised clinical predictors in the same files are excluded because they are trained on ClinVar.

Families are MMseqs2 [31] clusters of wild-type sequences at 30% identity and 80% bidirectional coverage: 2,081 families, of which 1,805 are single proteins (largest: 10 proteins) and 1,309 have at least ten variants. As sensitivity checks we use 50% identity (2,275 families) and single-linkage components of significant homology ($E \leq 10^{-3}$, identity $\geq 30\%$, both sequences covered $\geq 50\%$; 1,939 families), which merges homologs that share a domain. Family descriptors are computed without labels: maximum identity to any training protein over significant MMseqs2 hits, $\log N_{\mathrm{eff}}$ and N_{eff} per position from ProteinGym's alignment weights, cosine distance of mean-pooled ESM2-35M [21] embeddings to the nearest training protein, log length, log number of scored variants, and, for one ablation, summaries of the predictor's scores. A family is an *orphan* if no member has a significant hit at $\geq 30\%$ identity outside the family (1,276 families).

6.2 Protocol

Each of ten seeds splits families into 40% training, 30% calibration and 30% test (832, 624 and 625 families). **S1** draws all three uniformly at random. **S2** keeps the S1 training families and samples test families without replacement with probability proportional to $\exp(-\beta z)$, where z is the standardized mean of maximum identity to training and $\log N_{\mathrm{eff}}$, so that test families are poorly supported and calibration families well supported; $\beta \in \{0.5, 1, 2\}$ ($\beta = 0$ is S1). At $\beta = 1$, mean maximum identity to training is 0.105 for test and 0.221 for calibration families, and mean $\log N_{\mathrm{eff}}$ is 5.88 against 6.93. **S3** draws test families only from orphan families, a natural shift with no explicit model (mean maximum identity 0.048 against 0.236). **S4** holds out

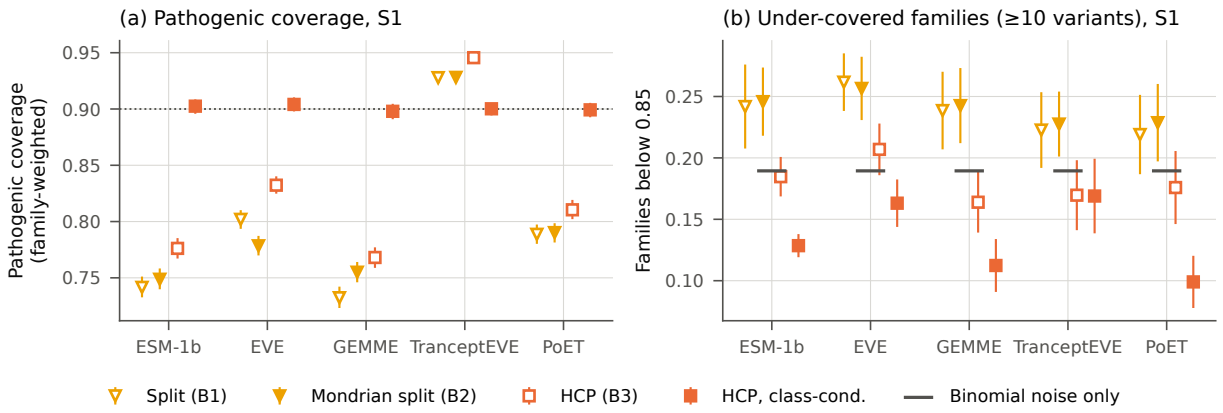


Fig. 1. Exchangeable families (S1, $\alpha = 0.1$). (a) Family-weighted pathogenic coverage Cov_1 with 95% family-bootstrap intervals; the dotted line is the target. (b) Fraction of families with at least ten variants whose overall coverage (both classes, for every method) is below 0.85 (mean and s.d. over seeds); the gray dash is the expected fraction if every family had coverage exactly 0.90.

regions of protein space: k-means groups families into 20 regions by the leading principal components of their mean ESM2 embeddings, each seed takes whole regions in random order as test families, and training and calibration families come from the remaining regions. Unlike S2, this shift is defined in embedding space rather than by the scalar descriptors that $\hat{w}$ uses; held-out families are farther from training in embedding distance (0.040 against 0.029) and have deeper alignments (mean $\log N_{\mathrm{eff}}$ 6.53 against 6.38).

The base model, fitted on training families only, is a logistic regression on the raw score (clipped to the training 0.1% and 99.9% quantiles) and the score standardized within its protein; its test AUC is 0.873–0.929 across predictors. Baselines are variant-level split conformal (B1) and its class-conditional version (B2); family-level HCP pooling and Dunn et al.'s single and repeated subsampling (B3) [18,10]; group-weighted conformal (B4) [3] with nine bins (maximum identity zero, low or high, crossed with $\log N_{\mathrm{eff}}$ terciles) and target proportions from the test families; and support-only and random abstention at ProtCOVER's abstention rate (B5). Every method has marginal and class-conditional versions. ProtCOVER uses 416 of the calibration families in the quantile and the rest to fit $\hat{w}$. The gate's kernel scale is chosen by inner risk-coverage AUC and $(m, \pi_{\max})$ as the least-abstaining setting within 0.005 of the best retained coverage under a pre-declared 10% abstention budget, all on a shifted ($\beta = 1$) inner split of the training families. Test families are used only for final evaluation, and only their label-free descriptors enter $\hat{w}$. Appendix B lists all remaining settings.

We report family-weighted coverage Cov, class-conditional Cov_1 (pathogenic) and Cov_0 (benign), variant-weighted coverage, mean set size, singleton rate, the fraction of families with at least ten variants whose coverage is below $1 - \alpha - 0.05$ (with its expectation under pure binomial noise), abstention rates, and risk-coverage curves (error of retained singleton sets against the retained fraction, abstaining whole families) with their area (AURC). Intervals are 95% bootstrap intervals that resample test families within each seed and average over seeds; differences between methods use the same resamples (paired). Unless stated otherwise $\alpha = 0.1$; results at $\alpha \in \{0.05, 0.2\}$ are in Appendix C.

6.3 Results

Variant-level calibration misses pathogenic variants per family. In S1 (Fig. 1), variant-level split conformal has variant-weighted coverage 0.894–0.901, close to its target, but family-weighted coverage 0.880–0.885; family-level HCP restores 0.900–0.901. The gap is concentrated in the pathogenic class: B2, calibrated class by class over variants, attains variant-weighted pathogenic coverage 0.893–0.905 for four predictors yet covers only 0.749–0.790 of pathogenic variants per family. The 5% of test families with the most pathogenic variants hold 50% of them, and their pathogenic variants are easier: under B2, families with fewer than five pathogenic variants have pathogenic coverage 0.660–0.727 against 0.899–0.921 for families with at least twenty. TranceptEVE is the exception (0.928), with the error moved to the benign class. Family-level class-conditional

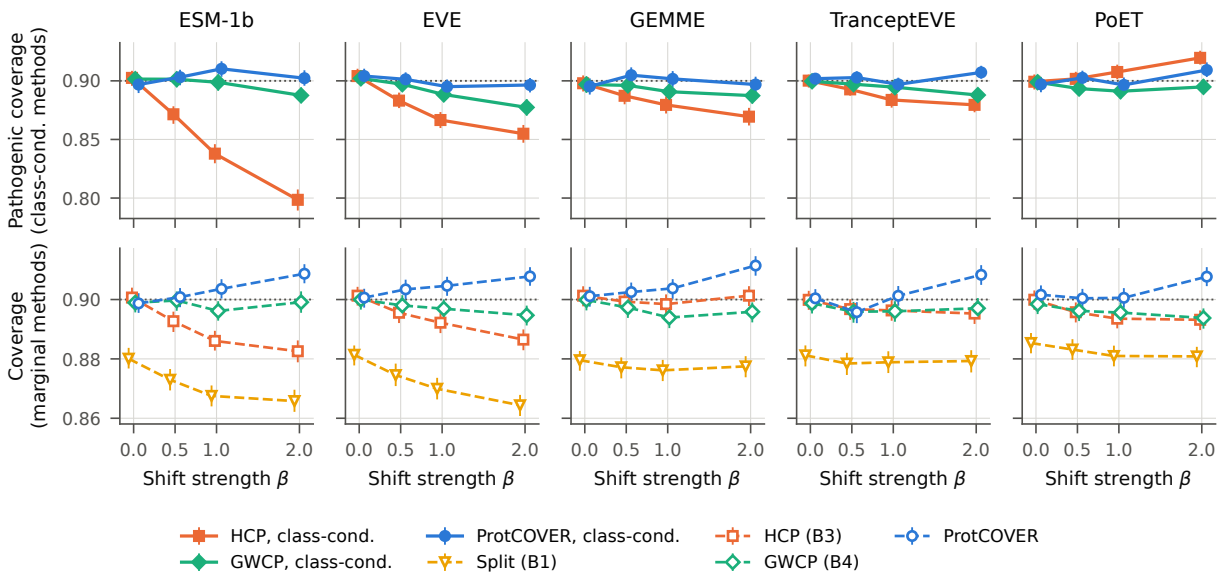


Fig. 2. Coverage under increasing descriptor shift (S2; $\beta = 0$ is S1), $\alpha = 0.1$, with 95% family-bootstrap intervals; methods are offset slightly in β. Top: pathogenic coverage Cov_1 of class-conditional family-level methods (Mondrian split, far below the target for four predictors, is in Table 1). Bottom: family-weighted coverage Cov of marginal methods.

calibration attains 0.898–0.904 (pathogenic) and 0.898–0.903 (benign), as Propositions 1 and 2 guarantee, and lowers the fraction of under-covered families below the binomial-noise level (Fig. 1b), at the cost of larger sets.

Weighting restores coverage under family-level shift. Under S2 (Fig. 2, Table 1), family-level calibration alone degrades as β grows: at $\beta = 1$ its pathogenic coverage spans 0.838–0.908 across predictors, and for ESM-1b it falls to 0.798 at $\beta = 2$. Its error is not one-sided: for PoET, and for TranceptEVE in S3, it over-covers pathogenic variants while under-covering benign ones. We therefore score each method by the shortfall of its worse class below the target. Over the 25 shifted predictor-setting cells (S2 with $\beta > 0$, and S3), this shortfall averages 0.023 for HCP (maximum 0.102; worse class below 0.89 in 17 cells), 0.007 for GWCP (0.027; 7 cells), and 0.003 for ProtCOVER (0.014; 2 cells). Both components matter: the same weights applied to variant-level calibration leave a mean shortfall of 0.122 (Table 2). GWCP, whose bins use the two descriptors that define the S2 shift, is the strongest baseline; at $\beta = 2$ ProtCOVER's pathogenic coverage exceeds it by +0.010 to +0.019 with paired intervals above zero for all five predictors (Appendix Table 3). S2 favors weighting by construction, since it shifts the very descriptors that $\hat{w}$ models. On the natural shift of S3, ProtCOVER only matches GWCP (differences −0.003 to +0.005), while still correcting HCP's errors in both directions. On held-out regions of embedding space (S4), variant-level Mondrian calibration again covers only 0.732–0.798 of pathogenic variants per family for four predictors, but family-level calibration alone is already close to the target (worse-class shortfall 0.006, maximum 0.013), and weighting brings no gain: ProtCOVER's shortfall is 0.008 (maximum 0.014) against 0.004 for GWCP. With little bias left to correct, the estimated weights mainly add variance; in S4, looser clipping lowers coverage (Table 2). The fraction of under-covered families is similar for the three family-level methods under shift (ProtCOVER 13%, GWCP 14%, HCP 14%; binomial-noise reference 20%). Our bootstrap intervals are conditional on the splits; across seeds, ProtCOVER's pathogenic coverage in a cell has standard deviation 0.013–0.044, so differences of about 0.01 between methods should be read with care. The price is larger sets (Fig. 3): averaged over the shifted cells, mean set size is 1.33 for ProtCOVER, 1.31 for GWCP and 1.30 for HCP, and ProtCOVER returns both labels for 29–44% of variants, depending on predictor and setting. Along the α curves, ProtCOVER's curve lies above HCP's for ESM-1b and EVE under S2; elsewhere the family-level curves nearly coincide, so the extra size mostly moves along a shared coverage-size trade-off toward the target rather than improving it.

Table 1. Family-weighted pathogenic / benign coverage (Cov_1 / Cov_0) of class-conditional methods at $\alpha = 0.1$ (target 0.90), mean over ten seeds. The ProtCOVER row omits the gate, which can only raise coverage (Lemma 1); the full method is in Table 2. Paired differences with intervals are in Table 3.

Setting	Method	ESM-1b	EVE	GEMME	TranceptEVE	PoET
S1 (exchangeable)	Mondrian split (B2)	0.749 / 0.899	0.778 / 0.898	0.755 / 0.895	0.928 / 0.837	0.790 / 0.877
	HCP (B3)	0.903 / 0.903	0.904 / 0.901	0.898 / 0.898	0.900 / 0.901	0.899 / 0.903
	GWCP (B4)	0.902 / 0.901	0.902 / 0.899	0.897 / 0.897	0.900 / 0.900	0.899 / 0.901
	ProtCOVER (i)+(ii)	0.897 / 0.904	0.904 / 0.902	0.895 / 0.899	0.902 / 0.902	0.897 / 0.906
S2, $\beta = 1$	Mondrian split (B2)	0.655 / 0.929	0.709 / 0.915	0.712 / 0.906	0.920 / 0.839	0.795 / 0.856
	HCP (B3)	0.838 / 0.927	0.867 / 0.916	0.879 / 0.909	0.884 / 0.894	0.908 / 0.875
	GWCP (B4)	0.899 / 0.904	0.888 / 0.909	0.891 / 0.904	0.895 / 0.894	0.891 / 0.897
	ProtCOVER (i)+(ii)	0.910 / 0.905	0.895 / 0.909	0.902 / 0.908	0.897 / 0.898	0.897 / 0.905
S2, $\beta = 2$	Mondrian split (B2)	0.619 / 0.941	0.686 / 0.917	0.700 / 0.910	0.921 / 0.839	0.806 / 0.840
	HCP (B3)	0.798 / 0.939	0.855 / 0.917	0.869 / 0.914	0.880 / 0.891	0.920 / 0.853
	GWCP (B4)	0.888 / 0.905	0.877 / 0.906	0.887 / 0.897	0.888 / 0.889	0.895 / 0.873
	ProtCOVER (i)+(ii)	0.902 / 0.915	0.896 / 0.916	0.897 / 0.915	0.907 / 0.899	0.909 / 0.891
S3 (orphan families)	Mondrian split (B2)	0.678 / 0.915	0.772 / 0.895	0.724 / 0.903	0.943 / 0.823	0.817 / 0.854
	HCP (B3)	0.876 / 0.908	0.909 / 0.899	0.890 / 0.905	0.914 / 0.892	0.931 / 0.879
	GWCP (B4)	0.889 / 0.905	0.903 / 0.898	0.895 / 0.900	0.903 / 0.913	0.901 / 0.900
	ProtCOVER (i)+(ii)	0.894 / 0.908	0.907 / 0.901	0.900 / 0.908	0.900 / 0.921	0.901 / 0.910
S4 (held-out regions)	Mondrian split (B2)	0.732 / 0.904	0.780 / 0.893	0.763 / 0.897	0.938 / 0.820	0.798 / 0.869
	HCP (B3)	0.891 / 0.899	0.904 / 0.900	0.907 / 0.898	0.908 / 0.887	0.910 / 0.895
	GWCP (B4)	0.896 / 0.902	0.897 / 0.902	0.904 / 0.895	0.900 / 0.895	0.899 / 0.901
	ProtCOVER (i)+(ii)	0.886 / 0.905	0.894 / 0.913	0.907 / 0.891	0.888 / 0.913	0.899 / 0.918

The support gate preserves coverage but does not improve selective risk. With $(m, \pi_{\max})$ and the kernel scale tuned on training families under a 10% budget, the gate abstained on at most 1.2% of test families except at $\beta = 2$ (0.2%–3.6%), so the full method matches (i)+(ii) (Table 2). Risk-coverage curves (Appendix Fig. 4) evaluate the ranking itself at every abstention level, against random abstention averaged over 100 random family orders. Abstaining first on the families with the lowest ESS gave a lower AURC than random abstention in the majority of seeds in 11 of 30 predictor-setting cells and a higher one in 13 (the rest tied); ranking by maximum identity (12 lower, 14 higher) or by test-point mass (11 lower, 15 higher) did no better. On this benchmark, how well a family is supported by calibration data does not predict how often the predictor errs on it once coverage is controlled. The gate therefore adds nothing measurable here; Lemma 1 guarantees only that it costs no coverage.

Ablations. Table 2 shows that sequence similarity or embedding distance alone does not correct the S2 shift, alignment depth alone helps on average but not for every predictor, and the support descriptors together match the full set; adding the predictor's score summaries does not help and lowers the worst predictor's coverage at $\beta = 2$. Embedding distance contributes little here because it barely differs between calibration and test families (mean 0.032 against 0.029 at $\beta = 1$). Tight clipping ($c = 5$) under-corrects at $\beta = 2$, while loose or no clipping hurts in S4, so the clipping level trades bias correction against variance. Under the alternative family definitions, ProtCOVER's mean pathogenic coverage over predictors stays at 0.899–0.906 across the shifted settings; its worst predictor-setting cell is 0.873 with 50% clustering and 0.887 with homology components, against 0.788 and 0.804 for HCP.

6.4 Synthetic validation

We simulate families with a descriptor $d \sim N(0, 1)$, random offsets, and slopes on the score whose mean depends on d, with a fixed base model, so that the truth is known (300 replicates, Monte Carlo standard error at most 0.001). Under A1 with larger families being easier, variant-level split conformal has variant-weighted coverage 0.900 but family-weighted coverage 0.831, while HCP attains 0.906 and single subsampling 0.902. Under exact family-level covariate shift (A2, test descriptors $N(-\beta, 1)$), HCP drops to 0.826 at $\beta = 1$ and 0.734 at $\beta = 2$; weighting with the true ratio attains 0.911 and 0.950, and with cross-fitted estimated ratios 0.916 and 0.951 (class-conditional pathogenic: 0.916 and 0.947). The excess at $\beta = 2$ is the price of the test-point mass, which grows when few calibration families resemble the test family. These checks are also unit tests in the released code.

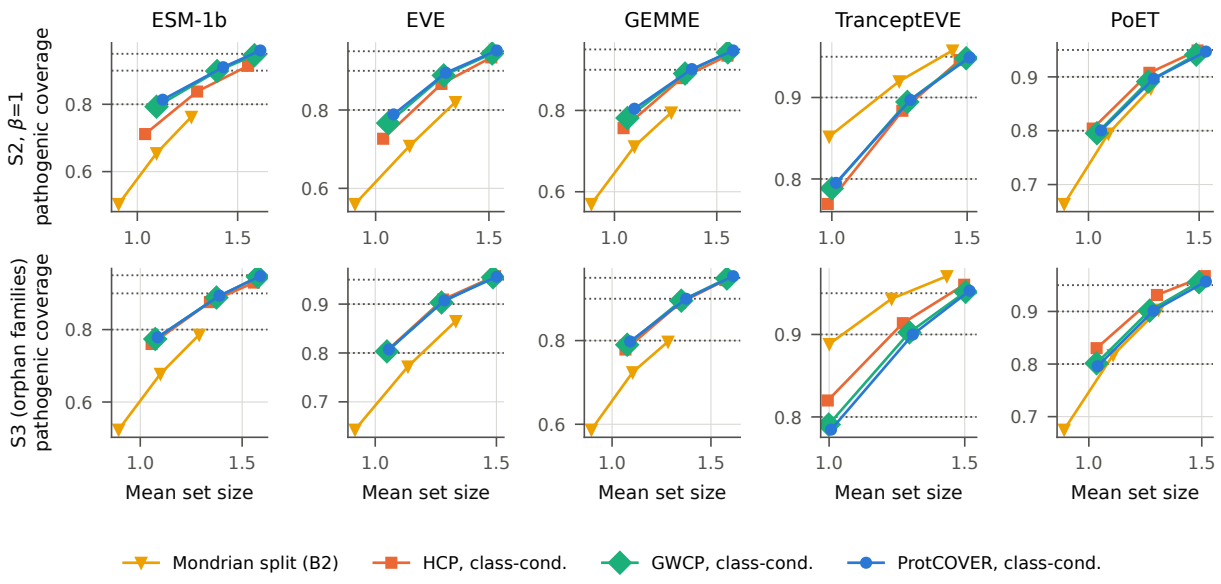


Fig. 3. Pathogenic coverage against mean set size for $\alpha \in \{0.2, 0.1, 0.05\}$ (left to right along each curve), class-conditional methods. Dotted lines mark the three targets.

7 Limitations

The guarantees are marginal over families: ProtCOVER does not certify coverage for any particular protein, and the per-family coverage of small families is dominated by binomial noise. The weighted guarantee assumes that, given its descriptors, a family's labels and scores follow the same law in calibration and deployment; our S2 splits satisfy this only approximately (finite-pool sampling), S3 is not constructed to satisfy it, and Corollary 1 bounds the effect of estimated weights by an unobservable distance. S2 shifts the very descriptors that $\hat{w}$ uses, which favors weighting; on the natural shift of S3, ProtCOVER only matches group-weighted conformal prediction, and on held-out embedding regions (S4), where family-level calibration alone suffices, its weights add variance without a gain. Deciding from label-free data alone when to weight is an open problem. The density ratio is transductive: it needs a batch of unlabeled target families, and a single query would require a reference panel, such as the unlabeled proteome, in their place. The 30%/80% clustering leaves most families as single proteins, and partial-coverage homologs can fall in different families; the stricter homology components did not change our conclusions, but deeper remote homology could still link training and test families. ClinVar labels are noisy and enriched in well-studied genes, every protein in the benchmark has both pathogenic and benign variants, and the base model's within-protein standardization and the variant-count descriptor are computed over the labeled variant set, so set sizes may differ for a typical query protein. The base model is deliberately simple, and a stronger one may give smaller sets. ProtCOVER outputs prediction sets, not the evidence strengths of ACMG/AMP; mapping sets to PP3/BP4 strengths is future work. Finally, the support gate did not improve selective risk here; whether support predicts error for other predictors, variant classes, or benchmarks remains open.

Table 2. Ablations: pathogenic coverage Cov_1 at $\alpha = 0.1$, mean (minimum) over the five predictors. For family definitions, HCP / ProtCOVER.

Configuration	S2, $\beta = 1$	S2, $\beta = 2$	S3	S4
Components				
(i) family-level calibration (HCP)	0.875 (0.838)	0.864 (0.798)	0.904 (0.876)	0.904 (0.891)
(i)+(ii) weighting	0.900 (0.895)	0.902 (0.896)	0.900 (0.894)	0.895 (0.886)
(i)+(iii) gate	0.875 (0.838)	0.864 (0.798)	0.904 (0.876)	0.904 (0.891)
(i)+(ii)+(iii) full	0.900 (0.895)	0.904 (0.898)	0.901 (0.894)	0.895 (0.886)
(ii) without (i): variant-level weighting	0.789 (0.740)	0.787 (0.723)	0.807 (0.740)	0.798 (0.737)
Descriptors for $\hat{w}$				
sequence similarity only	0.871 (0.850)	0.853 (0.799)	0.899 (0.892)	0.906 (0.903)
alignment depth only	0.899 (0.892)	0.901 (0.872)	0.899 (0.878)	0.903 (0.893)
embedding distance only	0.879 (0.845)	0.866 (0.806)	0.904 (0.872)	0.895 (0.882)
the three support groups	0.900 (0.894)	0.904 (0.898)	0.900 (0.894)	0.896 (0.887)
main set + predictor score summaries	0.903 (0.894)	0.898 (0.889)	0.902 (0.895)	0.894 (0.887)
Weight clipping c (default 20)				
$c = 5$	0.899 (0.894)	0.889 (0.870)	0.900 (0.892)	0.904 (0.897)
$c = 100$	0.900 (0.895)	0.908 (0.904)	0.901 (0.894)	0.883 (0.874)
no clipping	0.900 (0.895)	0.908 (0.904)	0.901 (0.894)	0.837 (0.779)
Family definition (main: MMseqs2 30% identity, 80% coverage)				
MMseqs2 50% identity: HCP	0.882 (0.835)	0.869 (0.788)	0.903 (0.872)	0.895 (0.879)
MMseqs2 50% identity: (i)+(ii)	0.903 (0.897)	0.903 (0.873)	0.906 (0.897)	0.899 (0.882)
homology components: HCP	0.881 (0.839)	0.871 (0.804)	0.910 (0.894)	0.903 (0.893)
homology components: (i)+(ii)	0.901 (0.896)	0.902 (0.892)	0.905 (0.901)	0.904 (0.887)

Code Availability

Code, configuration, and scripts that regenerate every number, table, and figure in this paper from the public ProteinGym release are available at `https://github.com/wael-fahmy/protcover` under the MIT license.

References

1. Angelopoulos, A.N., Bates, S.: Conformal prediction: A gentle introduction. Foundations and Trends in Machine Learning **16**(4), 494–591 (2023). https://doi.org/10.1561/2200000101
2. Barber, R.F., Candès, E.J., Ramdas, A., Tibshirani, R.J.: Conformal prediction beyond exchangeability. The Annals of Statistics **51**(2), 816–845 (2023). https://doi.org/10.1214/23-AOS2276
3. Bhattacharyya, A., Barber, R.F.: Group-weighted conformal prediction. arXiv preprint arXiv:2401.17452 (2024)
4. Bhattacharyya, A., Ding, T., Barber, R.F.: Conformal prediction with macro-coverage guarantees. arXiv preprint arXiv:2606.28598 (2026)
5. Boger, R.S., Chithrananda, S., Angelopoulos, A.N., Yoon, P.H., Jordan, M.I., Doudna, J.A.: Functional protein mining with conformal guarantees. Nature Communications **16**, 85 (2025). https://doi.org/10.1038/s41467-024-55676-y
6. Brandes, N., Goldman, G., Wang, C.H., Ye, C.J., Ntranos, V.: Genome-wide prediction of disease variant effects with a deep protein language model. Nature Genetics **55**, 1512–1522 (2023). https://doi.org/10.1038/s41588-023-01465-0
7. Cauchois, M., Gupta, S., Ali, A., Duchi, J.C.: Robust validation: Confident predictions even when distributions shift. Journal of the American Statistical Association (2024). https://doi.org/10.1080/01621459.2023.2298037
8. Chen, Y., Fayer, S., Jain, S., Benazouz, M., et al.: Gene- and domain-aware calibration increases the clinical utility of variant effect predictors. bioRxiv (2026). https://doi.org/10.64898/2026.02.17.706269
9. Ding, T., Angelopoulos, A.N., Bates, S., Jordan, M.I., Tibshirani, R.J.: Class-conditional conformal prediction with many classes. In: Advances in Neural Information Processing Systems. vol. 36 (2023), arXiv:2306.09335
10. Dunn, R., Wasserman, L., Ramdas, A.: Distribution-free prediction sets for two-layer hierarchical models. Journal of the American Statistical Association **118**(544), 2491–2502 (2023). https://doi.org/10.1080/01621459.2022.2060112
11. Fannjiang, C., Bates, S., Angelopoulos, A.N., Listgarten, J., Jordan, M.I.: Conformal prediction under feedback covariate shift for biomolecular design. Proceedings of the National Academy of Sciences **119**(43), e2204569119 (2022). https://doi.org/10.1073/pnas.2204569119
12. Frazer, J., Notin, P., Dias, M., Gomez, A., Min, J.K., Brock, K., Gal, Y., Marks, D.S.: Disease variant prediction with deep generative models of evolutionary data. Nature **599**, 91–95 (2021). https://doi.org/10.1038/s41586-021-04043-8
13. Geifman, Y., El-Yaniv, R.: Selective classification for deep neural networks. In: Advances in Neural Information Processing Systems. vol. 30 (2017), arXiv:1705.08500

14. Groth, P.M., Kerrn, M.H., Olsen, L., Salomon, J., Boomsma, W.: Kermut: Composite kernel regression for protein variant effects. In: Advances in Neural Information Processing Systems. vol. 37 (2024), arXiv:2407.00002
15. Guan, L.: Localized conformal prediction: A generalized inference framework for conformal prediction. Biometrika **110**(1), 33–50 (2023). https://doi.org/10.1093/biomet/asac040
16. Laine, E., Karami, Y., Carbone, A.: GEMME: A simple and fast global epistatic model predicting mutational effects. Molecular Biology and Evolution **36**(11), 2604–2619 (2019). https://doi.org/10.1093/molbev/msz179
17. Landrum, M.J., et al.: Clinvar: Improving access to variant interpretations and supporting evidence. Nucleic Acids Research **46**(D1), D1062–D1067 (2018). https://doi.org/10.1093/nar/gkx1153
18. Lee, Y., Barber, R.F., Willett, R.: Distribution-free inference with hierarchical data. ACM/IMS Journal of Data Science (2026). https://doi.org/10.1145/3786352
19. Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R.J., Wasserman, L.: Distribution-free predictive inference for regression. Journal of the American Statistical Association **113**(523), 1094–1111 (2018). https://doi.org/10.1080/01621459.2017.1307116
20. Lei, L., Candès, E.J.: Conformal inference of counterfactuals and individual treatment effects. Journal of the Royal Statistical Society Series B: Statistical Methodology **83**(5), 911–938 (2021). https://doi.org/10.1111/rssb.12445
21. Lin, Z., Akin, H., Rao, R., Hie, B., Zhu, Z., Lu, W., Smetanin, N., Verkuil, R., Kabeli, O., Shmueli, Y., dos Santos Costa, A., Fazel-Zarandi, M., Sercu, T., Candido, S., Rives, A.: Evolutionary-scale prediction of atomic-level protein structure with a language model. Science **379**(6637), 1123–1130 (2023). https://doi.org/10.1126/science.ade2574
22. Liu, Y., Levis, A.W., Normand, S.L., Han, L.: Multi-source conformal inference under distribution shift. In: Proceedings of the 41st International Conference on Machine Learning. PMLR, vol. 235 (2024), arXiv:2405.09331
23. Mallick, S., Tchetgen Tchetgen, E., Dobriban, E., Lee, Y.: Generalized hierarchical conformal prediction. arXiv preprint arXiv:2608.15500 (2026)
24. Notin, P., Kollasch, A.W., Ritter, D., van Niekerk, L., Paul, S., Spinner, H., Rollins, N., Shaw, A., Orenbuch, R., Weitzman, R., Frazer, J., Dias, M., Franceschi, D., Gal, Y., Marks, D.S.: Proteingym: Large-scale benchmarks for protein design and fitness prediction. bioRxiv (2023). https://doi.org/10.1101/2023.12.07.570727, also in Advances in Neural Information Processing Systems 36, Datasets and Benchmarks Track
25. Notin, P., Van Niekerk, L., Kollasch, A.W., Ritter, D., Gal, Y., Marks, D.S.: TranceptEVE: Combining family-specific and family-agnostic models of protein sequences for improved fitness prediction. bioRxiv (2022). https://doi.org/10.1101/2022.12.07.519495
26. Passi, A., Schneidman-Duhovny, D.: Calibrated variant effect prediction at the residue level using conditional score distributions. bioRxiv (2025). https://doi.org/10.1101/2025.11.24.690189
27. Pejaver, V., et al.: Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria. The American Journal of Human Genetics **109**(12), 2163–2177 (2022). https://doi.org/10.1016/j.ajhg.2022.10.013
28. Podkopaev, A., Ramdas, A.: Distribution-free uncertainty quantification for classification under label shift. In: Proceedings of the 37th Conference on Uncertainty in Artificial Intelligence. PMLR, vol. 161 (2021), arXiv:2103.03323
29. Richards, S., et al.: Standards and guidelines for the interpretation of sequence variants. Genetics in Medicine **17**(5), 405–424 (2015). https://doi.org/10.1038/gim.2015.30
30. Rives, A., Meier, J., Sercu, T., Goyal, S., Lin, Z., Liu, J., Guo, D., Ott, M., Zitnick, C.L., Ma, J., Fergus, R.: Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. Proceedings of the National Academy of Sciences **118**(15), e2016239118 (2021). https://doi.org/10.1073/pnas.2016239118
31. Steinegger, M., Söding, J.: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology **35**, 1026–1028 (2017). https://doi.org/10.1038/nbt.3988
32. Tejura, M., Fayer, S., McEwen, A.E., Flynn, J., et al.: Calibration of variant effect predictors on genome-wide data masks heterogeneous performance across genes. The American Journal of Human Genetics (2024). https://doi.org/10.1016/j.ajhg.2024.07.018
33. Tibshirani, R.J., Barber, R.F., Candès, E.J., Ramdas, A.: Conformal prediction under covariate shift. In: Advances in Neural Information Processing Systems. vol. 32 (2019), arXiv:1904.06019
34. Truong, Jr., T.F., Bepler, T.: PoET: A generative model of protein families as sequences-of-sequences. In: Advances in Neural Information Processing Systems. vol. 36 (2023), arXiv:2306.06156
35. Vovk, V.: Conditional validity of inductive conformal predictors. In: Proceedings of the Asian Conference on Machine Learning. PMLR, vol. 25, pp. 475–490 (2012), arXiv:1209.2673
36. Vovk, V., Gammerman, A., Shafer, G.: Algorithmic Learning in a Random World. Springer (2005). https://doi.org/10.1007/b106715

A Proofs

Throughout, $F_k = N_k^{-1} \sum_{i=1}^{N_k} \delta_{S_{k,i}}$ is the score distribution of family k (for the test family it involves unobserved labels and is used only in the analysis), and $\gamma_{K+1} = F_{K+1}((-\infty, \hat{q}])$ because a test variant is covered iff its score is at most $\hat{q}$.

A comparison step used in every proof. Fix weights $p_1, \dots, p_{K+1}$ and define the oracle measure $\mu = \sum_{k=1}^{K+1} p_k F_k$ and $q = \mathrm{Quant}_{1-\alpha}(\mu)$. The computable $\hat{\mu}$ of (2) is obtained from μ by moving the mass p_{K+1} from F_{K+1} to $+\infty$, so $\hat{\mu}((-\infty, u]) \le \mu((-\infty, u])$ for every u and hence $\hat{q} \ge q$. Consequently $\gamma_{K+1} = F_{K+1}((-\infty, \hat{q}]) \ge F_{K+1}((-\infty, q])$. Also $\mu((-\infty, q]) \ge 1 - \alpha$ by right-continuity of distribution functions.

Proof (Proposition 1). With $\omega \equiv 1$, $p_k = 1/(K+1)$ and q is a symmetric function of $(G_1, \dots, G_{K+1})$. By exchangeability, $\mathbb{E}[F_{K+1}((-\infty, q])] = \mathbb{E}[F_k((-\infty, q])]$ for every k, so

$$\mathbb{E}\big[F_{K+1}((-\infty, q])\big] = \mathbb{E}\Big[\frac{1}{K+1} \sum_{k=1}^{K+1} F_k((-\infty, q])\Big] = \mathbb{E}\big[\mu((-\infty, q])\big] \ge 1 - \alpha.$$

The comparison step gives $\mathrm{Cov}(C) = \mathbb{E}\gamma_{K+1} \ge 1 - \alpha$.

Proof (Theorem 1). Under A2 the joint law of $(G_1, \dots, G_{K+1})$ has density $\prod_{k \le K} f(g_k) \cdot w(d_{K+1}) f(g_{K+1})$ with respect to a product base measure, where f is a density of P; the sequence is therefore weighted exchangeable with weight functions $1, \dots, 1, w$ in the sense of [33, Definition 1 and Lemma 2]. Let E be the event that the unordered collection $\{G_1, \dots, G_{K+1}\}$ equals $\{g_1, \dots, g_{K+1}\}$. Exactly as in the proof of Lemma 3 of [33], $\mathbb{P}(G_{K+1} = g_k \mid E) = w(d_k) / \sum_j w(d_j) = p_k$ (summing p_k over coincident values, if any). The weights p_k and hence μ and q are functions of E alone, so

$$\mathbb{E}\big[F_{K+1}((-\infty, q]) \,\big|\, E\big] = \sum_{k=1}^{K+1} p_k \, F_k((-\infty, q]) = \mu((-\infty, q]) \ge 1 - \alpha.$$

Taking expectations and applying the comparison step yields $\mathrm{Cov}(C) \ge 1 - \alpha$. The weights in (2) are computable because they depend on the test family only through the label-free D_{K+1}.

Proof (Corollary 1). Condition on $\hat{w}$, which is independent of the families. The procedure is invariant to rescaling the weights, so it coincides with the procedure of Theorem 1 for the shift $\hat{Q}$, whose density ratio with respect to P is $\hat{w}/\mathbb{E}_P\hat{w}$. Let $\phi(g) \in [0,1]$ be the expected per-family coverage of a test family g over $G_1, \dots, G_K \sim P$, given $\hat{w}$ (the threshold depends on g only through its descriptor). Theorem 1 gives $\mathbb{E}_{\hat{Q}}\phi \ge 1 - \alpha$, and $|\mathbb{E}_Q\phi - \mathbb{E}_{\hat{Q}}\phi| \le d_{\mathrm{TV}}(Q, \hat{Q})$ for any $[0,1]$-valued ϕ. Hence $\mathbb{E}[\gamma_{K+1} \mid \hat{w}] = \mathbb{E}_Q\phi \ge 1 - \alpha - d_{\mathrm{TV}}(Q, \hat{Q})$, with $d_{\mathrm{TV}}(Q, \hat{Q}) = \frac{1}{2}\mathbb{E}_P|w(D) - \hat{w}(D)/\mathbb{E}_P\hat{w}(D)|$; taking expectations over $\hat{w}$ gives the unconditional bound.

Proof (Proposition 2). Fix y and let $\mathcal{I}_y = \{k \le K+1 : N_{k,y} > 0\}$. Under A1, conditionally on $\mathcal{I}_y$ and on $K+1 \in \mathcal{I}_y$, the families indexed by $\mathcal{I}_y$ are exchangeable, because the event $\{\mathcal{I}_y = I\}$ is invariant under permutations of I. Applying the proof of Proposition 1 to these families, with F_k replaced by the label-y score distribution F_k^y, gives $\mathrm{Cov}_y(C) \ge 1 - \alpha$. Under A2, conditioning each family on $N_{\cdot,y} > 0$ keeps the families independent, with laws $P_y = P(\cdot \mid N_y > 0)$ and $Q_y = Q(\cdot \mid N_y > 0)$ satisfying $dQ_y/dP_y \propto w(D)$ because $\mathbf{1}\{N_y > 0\}$ is a function of G. The proof of Theorem 1 then applies verbatim, and the normalizing constant cancels in p_k. For the corollary, $\hat{Q}_y = \hat{Q}(\cdot \mid N_y > 0)$ has $d\hat{Q}_y/dP_y \propto \hat{w}(D)$, and the same argument gives the bound with $d_{\mathrm{TV}}(Q_y, \hat{Q}_y)$.

Proof (Lemma 1). $\mathbf{1}\{Y \in C'(X)\} \ge \mathbf{1}\{Y \in C(X)\}$ pointwise, so every average and expectation of the indicators is at least as large for C'.

B Protocol Details

Predictors. The five unsupervised predictors are all the non-clinical models in ProteinGym v1.3's clinical score files (columns ESM1b, EVE, GEMME, TranceptEVE_L, PoET; PoET is the 200M model). The other columns are clinical predictors trained or tuned on clinical labels.

Families and homology. MMseqs2 release 18-8cc5c. Clusters: `cluster` with minimum identity 0.3 (0.5 for the sensitivity check), coverage 0.8 and coverage mode 0 (both sequences). Homology search: all against all, sensitivity 7.5, $E \leq 10$, at most 500 hits per query; hits are then filtered at $E \leq 10^{-3}$. Homology components additionally require identity ≥ 0.3 and coverage ≥ 0.5 of both sequences.

Descriptors. Per protein: maximum identity to any training protein over hits with $E \leq 10^{-3}$ (0 without a hit); $\log(1 + N_{\text{eff}})$ with N_{eff} the sum of ProteinGym's MSA sequence weights; log of N_{eff} divided by alignment length; cosine distance of the mean-pooled last-layer (layer 12) embedding of `esm2_t12_35M_UR50D` to the nearest training protein (sequences over 1022 residues are embedded in windows of 1022 with stride 511 and residue representations averaged over windows); log length. Per family: the maximum of maximum identity and the minimum of embedding distance over member proteins, the mean of the other protein descriptors, and log of the family's number of scored variants. Descriptors are recomputed for every split because they depend on its training families.

Splits. S2 uses z, the mean of the standardized maximum identity and $\log N_{\text{eff}}$ over non-training families, and Efraimidis–Spirakis keys $\log(u)\, e^{\beta z}$ with uniforms u shared across β, so the sweep is coupled and $\beta = 0$ is S1. S3 draws test families uniformly from orphan families and training and calibration families uniformly from the rest. S4 regions are fixed once, before any S4 run: each family's embedding is the mean of its members' unit-normalized ESM2 embeddings, renormalized; after centering, the first 50 principal components are clustered by k-means ($k = 20$, ten restarts, seed 0). Each seed permutes the regions and adds whole regions to the test set until it holds 30% of families (the last region is split at random); training and calibration families are drawn uniformly from the remaining families. The ten seeds resplit one fixed pool of families, so they are not independent samples of the population of families.

Base model and conformal methods. The base model standardizes its two features (clipped raw score and the score standardized over the protein's scored variants) and fits L_2-regularized logistic regression ($C = 1$). Dunn et al.'s repeated subsampling uses 50 subsamples. GWCP uses nine bins: maximum identity zero, at most the median of the positive values, or above it, crossed with $\log N_{\text{eff}}$ terciles over calibration and test families; its target proportions are the bin frequencies over all test families (label-free), and it pools at the family level like HCP. For $\hat{w}$, one third of the calibration families fit the classifier (logistic regression with ten candidate penalties, five-fold cross-validated log loss) and the test families are split into five folds.

Gate tuning. An inner split of each seed's training families holds out 40% as inner training families (descriptors are recomputed relative to them) and draws 30% as inner test families with the S2 sampler at $\beta = 1$; the rest are inner calibration families. Grids: kernel scale $\{0.1, 0.2, 0.35, 0.5, 1\}$ times the median pairwise distance of standardized support descriptors, $m \in \{0, 2, 5, 10, 20, 40\}$, $\pi_{\max} \in \{1, 0.05, 0.02, 0.01, 0.005\}$, abstention budget 10%, coverage tolerance 0.005.

Metrics. Coverage metrics average per-family coverage over families; set size and singleton rate average over variants. The under-covered fraction uses overall per-family coverage (both classes) for families with at least ten variants, for marginal and class-conditional methods alike; its binomial reference is the expected fraction under Binomial$(N_k, 1 - \alpha)$. Bootstrap intervals use 1000 resamples of test families within each seed, holding that seed's calibration draw fixed, so they are conditional on the splits; seed-to-seed variation is reported separately. Random abstention in risk-coverage curves is averaged over 100 random family orders per cell.

C Additional Results

Tables 4–6 report every method, marginal (cc = no) and class-conditional (cc = yes), averaged over the five predictors: family-weighted coverage Cov, variant-weighted coverage, Cov_1, Cov_0, mean set size, singleton rate, the fraction of families with at least ten variants whose coverage is below $1 - \alpha - 0.05$, and the fraction of test families on which the method abstained. Per-predictor values with bootstrap intervals are in the files `results/main/summary.csv` and `results/main/seed_summary.csv` of the code repository.

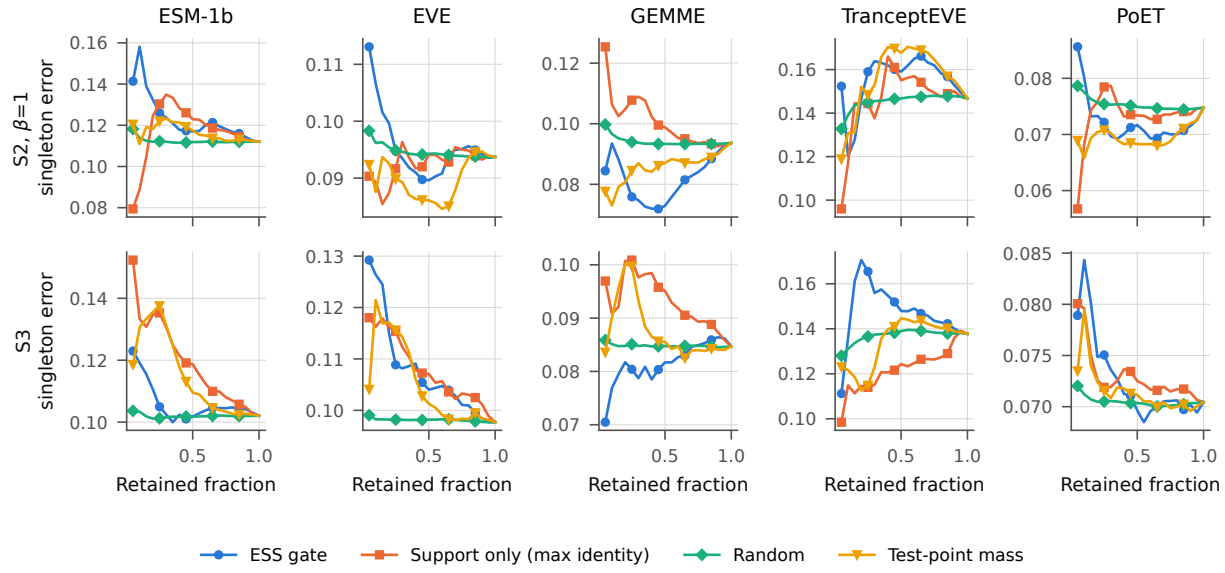


Fig. 4. Risk-coverage of abstention rules applied to ProtCOVER's class-conditional sets ($\alpha = 0.1$, mean over seeds): error of retained singleton sets as whole families are abstained in order of the score. Random abstention is averaged over 100 random family orders.

Table 3. Paired differences in family-weighted pathogenic (Cov_1) and benign (Cov_0) coverage between ProtCOVER (i)+(ii) and HCP or GWCP, class-conditional versions, $\alpha = 0.1$, mean over seeds, with 95% paired family-bootstrap intervals conditional on the splits.

		Cov_1 (pathogenic)		Cov_0 (benign)	
Setting	Predictor	− HCP	− GWCP	− HCP	− GWCP
S1	ESM-1b	−0.006 [−0.008, −0.004]	−0.005 [−0.006, −0.003]	+0.001 [+0.000, +0.002]	+0.003 [+0.002, +0.004]
	EVE	+0.000 [−0.001, +0.001]	+0.002 [+0.001, +0.003]	+0.001 [−0.000, +0.002]	+0.003 [+0.002, +0.004]
	GEMME	−0.003 [−0.004, −0.001]	−0.002 [−0.003, +0.000]	+0.000 [−0.000, +0.001]	+0.002 [+0.001, +0.003]
	TranceptEVE	+0.002 [+0.001, +0.003]	+0.002 [+0.001, +0.003]	+0.001 [+0.000, +0.002]	+0.002 [+0.002, +0.003]
	PoET	−0.002 [−0.004, −0.001]	−0.002 [−0.003, −0.000]	+0.003 [+0.002, +0.004]	+0.005 [+0.004, +0.006]
S2, $\beta = 1$	ESM-1b	+0.072 [+0.067, +0.078]	+0.012 [+0.009, +0.014]	−0.022 [−0.024, −0.021]	+0.000 [−0.001, +0.001]
	EVE	+0.028 [+0.025, +0.032]	+0.007 [+0.005, +0.009]	−0.007 [−0.008, −0.006]	+0.001 [−0.000, +0.002]
	GEMME	+0.022 [+0.019, +0.025]	+0.011 [+0.009, +0.013]	−0.001 [−0.003, −0.000]	+0.004 [+0.003, +0.005]
	TranceptEVE	+0.013 [+0.011, +0.016]	+0.002 [+0.001, +0.004]	+0.005 [+0.004, +0.006]	+0.004 [+0.003, +0.005]
	PoET	−0.011 [−0.013, −0.009]	+0.005 [+0.003, +0.008]	+0.030 [+0.028, +0.032]	+0.008 [+0.006, +0.009]
S2, $\beta = 2$	ESM-1b	+0.104 [+0.098, +0.111]	+0.015 [+0.011, +0.019]	−0.024 [−0.026, −0.022]	+0.010 [+0.009, +0.012]
	EVE	+0.042 [+0.037, +0.046]	+0.019 [+0.016, +0.022]	−0.001 [−0.003, +0.000]	+0.009 [+0.008, +0.011]
	GEMME	+0.027 [+0.024, +0.031]	+0.010 [+0.006, +0.013]	+0.001 [−0.001, +0.003]	+0.018 [+0.016, +0.020]
	TranceptEVE	+0.028 [+0.025, +0.030]	+0.019 [+0.017, +0.022]	+0.008 [+0.007, +0.010]	+0.010 [+0.009, +0.012]
	PoET	−0.010 [−0.013, −0.008]	+0.014 [+0.011, +0.018]	+0.038 [+0.036, +0.041]	+0.018 [+0.016, +0.021]
S3	ESM-1b	+0.018 [+0.015, +0.020]	+0.005 [+0.003, +0.007]	−0.000 [−0.002, +0.001]	+0.003 [+0.002, +0.004]
	EVE	−0.002 [−0.004, −0.000]	+0.004 [+0.002, +0.006]	+0.003 [+0.002, +0.004]	+0.003 [+0.002, +0.004]
	GEMME	+0.010 [+0.008, +0.012]	+0.005 [+0.003, +0.006]	+0.003 [+0.002, +0.004]	+0.008 [+0.006, +0.009]
	TranceptEVE	−0.014 [−0.016, −0.012]	−0.003 [−0.005, −0.001]	+0.029 [+0.027, +0.032]	+0.008 [+0.007, +0.010]
	PoET	−0.030 [−0.033, −0.027]	−0.000 [−0.002, +0.002]	+0.031 [+0.028, +0.034]	+0.009 [+0.008, +0.011]
S4	ESM-1b	−0.005 [−0.009, −0.001]	−0.010 [−0.013, −0.007]	+0.006 [+0.004, +0.008]	+0.004 [+0.002, +0.006]
	EVE	−0.010 [−0.014, −0.007]	−0.003 [−0.006, −0.000]	+0.013 [+0.011, +0.014]	+0.011 [+0.009, +0.012]
	GEMME	+0.000 [−0.003, +0.003]	+0.003 [+0.000, +0.006]	−0.007 [−0.009, −0.005]	−0.004 [−0.006, −0.002]
	TranceptEVE	−0.019 [−0.022, −0.016]	−0.011 [−0.014, −0.009]	+0.026 [+0.023, +0.028]	+0.018 [+0.016, +0.021]
	PoET	−0.011 [−0.015, −0.008]	−0.000 [−0.003, +0.003]	+0.023 [+0.020, +0.026]	+0.017 [+0.014, +0.019]

Table 4. All methods at $\alpha = 0.1$, mean over predictors and seeds.

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
S1 (exchangeable)	Split (B1/B2)	no	0.882	0.899	0.798	0.884	1.143	0.857	0.237	0.000
	Split (B1/B2)	yes	0.881	0.898	0.800	0.881	1.142	0.858	0.240	0.000
	HCP (B3)	no	0.901	0.918	0.827	0.903	1.204	0.796	0.180	0.000
	HCP (B3)	yes	0.922	0.929	0.901	0.901	1.304	0.696	0.134	0.000
	Subsample once (B3)	no	0.900	0.918	0.828	0.902	1.204	0.796	0.180	0.000
	Subsample once (B3)	yes	0.922	0.929	0.901	0.900	1.305	0.695	0.136	0.000
	Repeated subsample (B3)	no	0.901	0.918	0.826	0.903	1.205	0.795	0.180	0.000
	Repeated subsample (B3)	yes	0.923	0.929	0.901	0.901	1.305	0.695	0.134	0.000
	GWCP (B4)	no	0.899	0.917	0.824	0.901	1.199	0.801	0.185	0.000
	GWCP (B4)	yes	0.921	0.928	0.900	0.899	1.300	0.700	0.136	0.000
	ProtCOVER (i)+(ii)	no	0.900	0.918	0.826	0.903	1.204	0.796	0.181	0.000
	ProtCOVER (i)+(ii)	yes	0.923	0.930	0.899	0.902	1.303	0.697	0.133	0.000
	ProtCOVER (i)+(iii)	no	0.901	0.918	0.827	0.903	1.204	0.796	0.180	0.000
	ProtCOVER (i)+(iii)	yes	0.923	0.929	0.901	0.901	1.304	0.696	0.134	0.000
	ProtCOVER full	no	0.901	0.918	0.826	0.903	1.204	0.796	0.180	0.000
	ProtCOVER full	yes	0.923	0.930	0.899	0.903	1.304	0.696	0.133	0.000
	Support-only abst. (B5)	no	0.901	0.918	0.826	0.903	1.204	0.796	0.181	0.000
	Support-only abst. (B5)	yes	0.923	0.930	0.899	0.902	1.304	0.696	0.133	0.000
	Random abst. (B5)	no	0.901	0.918	0.826	0.903	1.204	0.796	0.180	0.000
	Random abst. (B5)	yes	0.923	0.930	0.899	0.902	1.304	0.696	0.133	0.000
S2, $\beta = 1$	Split (B1/B2)	no	0.875	0.893	0.772	0.883	1.137	0.863	0.252	0.000
	Split (B1/B2)	yes	0.873	0.891	0.758	0.889	1.137	0.863	0.255	0.000
	HCP (B3)	no	0.893	0.912	0.800	0.901	1.195	0.805	0.200	0.000
	HCP (B3)	yes	0.917	0.925	0.875	0.904	1.291	0.709	0.147	0.000
	Subsample once (B3)	no	0.893	0.912	0.800	0.902	1.195	0.805	0.200	0.000
	Subsample once (B3)	yes	0.918	0.926	0.877	0.904	1.294	0.706	0.146	0.000
	Repeated subsample (B3)	no	0.893	0.912	0.800	0.901	1.195	0.805	0.200	0.000
	Repeated subsample (B3)	yes	0.917	0.925	0.875	0.904	1.290	0.710	0.147	0.000
	GWCP (B4)	no	0.896	0.914	0.805	0.903	1.202	0.798	0.195	0.000
	GWCP (B4)	yes	0.921	0.928	0.893	0.902	1.318	0.682	0.140	0.000
	ProtCOVER (i)+(ii)	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	ProtCOVER (i)+(ii)	yes	0.925	0.931	0.900	0.905	1.340	0.660	0.130	0.000
	ProtCOVER (i)+(iii)	no	0.893	0.912	0.800	0.901	1.195	0.805	0.200	0.000
	ProtCOVER (i)+(iii)	yes	0.917	0.925	0.875	0.904	1.291	0.709	0.147	0.000
	ProtCOVER full	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	ProtCOVER full	yes	0.925	0.931	0.900	0.905	1.340	0.660	0.129	0.000
	Support-only abst. (B5)	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	Support-only abst. (B5)	yes	0.925	0.931	0.900	0.905	1.341	0.659	0.130	0.000
	Random abst. (B5)	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	Random abst. (B5)	yes	0.925	0.931	0.900	0.905	1.341	0.659	0.130	0.000
S2, $\beta = 2$	Split (B1/B2)	no	0.874	0.892	0.764	0.883	1.136	0.864	0.255	0.000
	Split (B1/B2)	yes	0.871	0.890	0.746	0.889	1.136	0.864	0.259	0.000
	HCP (B3)	no	0.892	0.911	0.793	0.900	1.192	0.808	0.204	0.000
	HCP (B3)	yes	0.914	0.923	0.864	0.903	1.283	0.717	0.156	0.000
	Subsample once (B3)	no	0.889	0.909	0.792	0.896	1.187	0.813	0.211	0.000
	Subsample once (B3)	yes	0.912	0.922	0.863	0.901	1.279	0.721	0.159	0.000
	Repeated subsample (B3)	no	0.891	0.911	0.792	0.900	1.192	0.808	0.204	0.000
	Repeated subsample (B3)	yes	0.914	0.923	0.864	0.903	1.283	0.717	0.156	0.000
	GWCP (B4)	no	0.896	0.915	0.801	0.904	1.212	0.788	0.196	0.000
	GWCP (B4)	yes	0.915	0.923	0.887	0.894	1.315	0.685	0.159	0.000
	ProtCOVER (i)+(ii)	no	0.909	0.927	0.826	0.914	1.260	0.740	0.161	0.000
	ProtCOVER (i)+(ii)	yes	0.927	0.935	0.902	0.907	1.363	0.637	0.127	0.000
	ProtCOVER (i)+(iii)	no	0.892	0.911	0.793	0.900	1.192	0.808	0.204	0.000
	ProtCOVER (i)+(iii)	yes	0.914	0.923	0.864	0.903	1.283	0.717	0.156	0.000
	ProtCOVER full	no	0.909	0.927	0.827	0.914	1.264	0.736	0.161	0.004
	ProtCOVER full	yes	0.928	0.936	0.904	0.908	1.372	0.628	0.125	0.022
	Support-only abst. (B5)	no	0.909	0.927	0.827	0.914	1.263	0.737	0.160	0.004
	Support-only abst. (B5)	yes	0.928	0.936	0.904	0.909	1.375	0.625	0.125	0.022
	Random abst. (B5)	no	0.909	0.928	0.827	0.915	1.264	0.736	0.160	0.004
	Random abst. (B5)	yes	0.928	0.936	0.904	0.909	1.375	0.625	0.125	0.022
S3 (orphan families)	Split (B1/B2)	no	0.875	0.891	0.789	0.878	1.137	0.863	0.254	0.000
	Split (B1/B2)	yes	0.874	0.890	0.787	0.878	1.135	0.865	0.251	0.000
	HCP (B3)	no	0.897	0.913	0.821	0.899	1.204	0.796	0.193	0.000
	HCP (B3)	yes	0.922	0.931	0.904	0.896	1.308	0.692	0.129	0.000
	Subsample once (B3)	no	0.896	0.912	0.820	0.898	1.204	0.796	0.195	0.000
	Subsample once (B3)	yes	0.922	0.931	0.904	0.896	1.309	0.691	0.128	0.000
	Repeated subsample (B3)	no	0.896	0.912	0.821	0.899	1.203	0.797	0.194	0.000
	Repeated subsample (B3)	yes	0.922	0.931	0.904	0.896	1.308	0.692	0.129	0.000
	GWCP (B4)	no	0.901	0.917	0.827	0.903	1.220	0.780	0.182	0.000
	GWCP (B4)	yes	0.924	0.931	0.898	0.903	1.312	0.688	0.128	0.000
	ProtCOVER (i)+(ii)	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	ProtCOVER (i)+(ii)	yes	0.928	0.934	0.900	0.910	1.329	0.671	0.120	0.000
	ProtCOVER (i)+(iii)	no	0.897	0.913	0.821	0.899	1.204	0.796	0.193	0.000
	ProtCOVER (i)+(iii)	yes	0.922	0.931	0.904	0.897	1.309	0.691	0.129	0.003
	ProtCOVER full	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	ProtCOVER full	yes	0.928	0.934	0.901	0.910	1.331	0.669	0.120	0.004
	Support-only abst. (B5)	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	Support-only abst. (B5)	yes	0.928	0.934	0.901	0.910	1.331	0.669	0.120	0.004
	Random abst. (B5)	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	Random abst. (B5)	yes	0.928	0.934	0.901	0.910	1.331	0.669	0.120	0.004

Table 5. All methods at $\alpha = 0.05$, mean over predictors and seeds.

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
S1 (exchangeable)	Split (B1/B2)	no	0.933	0.949	0.876	0.934	1.336	0.664	0.152	0.000
	Split (B1/B2)	yes	0.933	0.949	0.876	0.936	1.335	0.665	0.151	0.000
	HCP (B3)	no	0.950	0.966	0.904	0.953	1.443	0.557	0.095	0.000
	HCP (B3)	yes	0.963	0.968	0.952	0.950	1.517	0.483	0.075	0.000
	Subsample once (B3)	no	0.950	0.965	0.902	0.952	1.441	0.559	0.098	0.000
	Subsample once (B3)	yes	0.963	0.967	0.951	0.950	1.515	0.485	0.076	0.000
	Repeated subsample (B3)	no	0.951	0.966	0.904	0.953	1.444	0.556	0.094	0.000
	Repeated subsample (B3)	yes	0.963	0.968	0.952	0.950	1.517	0.483	0.076	0.000
	GWCP (B4)	no	0.949	0.964	0.901	0.951	1.432	0.568	0.101	0.000
	GWCP (B4)	yes	0.961	0.966	0.950	0.948	1.507	0.493	0.080	0.000
	ProtCOVER (i)+(ii)	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	ProtCOVER (i)+(ii)	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
	ProtCOVER (i)+(iii)	no	0.950	0.966	0.904	0.953	1.443	0.557	0.095	0.000
	ProtCOVER (i)+(iii)	yes	0.963	0.968	0.952	0.950	1.517	0.483	0.075	0.000
	ProtCOVER full	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	ProtCOVER full	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
	Support-only abst. (B5)	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	Support-only abst. (B5)	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
	Random abst. (B5)	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	Random abst. (B5)	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
S2, $\beta = 1$	Split (B1/B2)	no	0.928	0.946	0.859	0.934	1.330	0.670	0.167	0.000
	Split (B1/B2)	yes	0.927	0.944	0.843	0.940	1.325	0.675	0.167	0.000
	HCP (B3)	no	0.946	0.962	0.889	0.950	1.431	0.569	0.108	0.000
	HCP (B3)	yes	0.960	0.967	0.936	0.952	1.516	0.484	0.080	0.000
	Subsample once (B3)	no	0.947	0.963	0.890	0.951	1.441	0.559	0.105	0.000
	Subsample once (B3)	yes	0.958	0.966	0.935	0.949	1.510	0.490	0.084	0.000
	Repeated subsample (B3)	no	0.946	0.962	0.889	0.950	1.432	0.568	0.109	0.000
	Repeated subsample (B3)	yes	0.960	0.967	0.935	0.952	1.515	0.485	0.080	0.000
	GWCP (B4)	no	0.947	0.964	0.892	0.951	1.447	0.553	0.104	0.000
	GWCP (B4)	yes	0.961	0.966	0.944	0.950	1.527	0.473	0.081	0.000
	ProtCOVER (i)+(ii)	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	ProtCOVER (i)+(ii)	yes	0.965	0.969	0.951	0.955	1.553	0.447	0.069	0.000
	ProtCOVER (i)+(iii)	no	0.946	0.962	0.889	0.950	1.431	0.569	0.108	0.000
	ProtCOVER (i)+(iii)	yes	0.960	0.967	0.936	0.952	1.516	0.484	0.080	0.000
	ProtCOVER full	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	ProtCOVER full	yes	0.965	0.969	0.951	0.955	1.553	0.447	0.069	0.000
	Support-only abst. (B5)	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	Support-only abst. (B5)	yes	0.965	0.969	0.951	0.955	1.554	0.446	0.069	0.000
	Random abst. (B5)	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	Random abst. (B5)	yes	0.965	0.969	0.951	0.955	1.554	0.446	0.069	0.000
S2, $\beta = 2$	Split (B1/B2)	no	0.927	0.945	0.852	0.933	1.329	0.671	0.170	0.000
	Split (B1/B2)	yes	0.925	0.943	0.833	0.941	1.324	0.676	0.173	0.000
	HCP (B3)	no	0.944	0.961	0.884	0.949	1.427	0.573	0.114	0.000
	HCP (B3)	yes	0.957	0.966	0.929	0.950	1.512	0.488	0.087	0.000
	Subsample once (B3)	no	0.943	0.959	0.878	0.949	1.416	0.584	0.119	0.000
	Subsample once (B3)	yes	0.956	0.964	0.926	0.949	1.505	0.495	0.090	0.000
	Repeated subsample (B3)	no	0.945	0.961	0.883	0.949	1.427	0.573	0.115	0.000
	Repeated subsample (B3)	yes	0.957	0.966	0.929	0.949	1.511	0.489	0.087	0.000
	GWCP (B4)	no	0.946	0.963	0.888	0.950	1.456	0.544	0.109	0.000
	GWCP (B4)	yes	0.956	0.963	0.943	0.942	1.526	0.474	0.096	0.000
	ProtCOVER (i)+(ii)	no	0.962	0.975	0.924	0.963	1.572	0.428	0.068	0.000
	ProtCOVER (i)+(ii)	yes	0.969	0.974	0.962	0.958	1.612	0.388	0.064	0.000
	ProtCOVER (i)+(iii)	no	0.944	0.961	0.884	0.949	1.427	0.573	0.114	0.000
	ProtCOVER (i)+(iii)	yes	0.957	0.966	0.929	0.950	1.512	0.488	0.087	0.000
	ProtCOVER full	no	0.962	0.975	0.924	0.963	1.572	0.428	0.068	0.000
	ProtCOVER full	yes	0.969	0.974	0.962	0.958	1.613	0.387	0.064	0.014
	Support-only abst. (B5)	no	0.962	0.975	0.924	0.963	1.572	0.428	0.068	0.000
	Support-only abst. (B5)	yes	0.969	0.974	0.962	0.958	1.617	0.383	0.063	0.014
	Random abst. (B5)	no	0.962	0.975	0.924	0.963	1.572	0.428	0.068	0.000
	Random abst. (B5)	yes	0.969	0.974	0.962	0.958	1.616	0.384	0.063	0.014
S3 (orphan families)	Split (B1/B2)	no	0.929	0.944	0.870	0.931	1.330	0.670	0.159	0.000
	Split (B1/B2)	yes	0.930	0.944	0.865	0.936	1.330	0.670	0.156	0.000
	HCP (B3)	no	0.949	0.963	0.903	0.951	1.448	0.552	0.096	0.000
	HCP (B3)	yes	0.963	0.970	0.952	0.950	1.527	0.473	0.069	0.000
	Subsample once (B3)	no	0.946	0.961	0.899	0.948	1.440	0.560	0.105	0.000
	Subsample once (B3)	yes	0.962	0.970	0.952	0.948	1.525	0.475	0.070	0.000
	Repeated subsample (B3)	no	0.949	0.963	0.903	0.951	1.447	0.553	0.097	0.000
	Repeated subsample (B3)	yes	0.963	0.970	0.952	0.950	1.528	0.472	0.068	0.000
	GWCP (B4)	no	0.951	0.965	0.905	0.954	1.467	0.533	0.091	0.000
	GWCP (B4)	yes	0.964	0.969	0.951	0.953	1.527	0.473	0.069	0.000
	ProtCOVER (i)+(ii)	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	ProtCOVER (i)+(ii)	yes	0.967	0.971	0.953	0.958	1.548	0.452	0.063	0.000
	ProtCOVER (i)+(iii)	no	0.949	0.963	0.903	0.951	1.448	0.552	0.096	0.000
	ProtCOVER (i)+(iii)	yes	0.963	0.970	0.952	0.950	1.527	0.473	0.069	0.000
	ProtCOVER full	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	ProtCOVER full	yes	0.967	0.971	0.953	0.958	1.548	0.452	0.063	0.000
	Support-only abst. (B5)	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	Support-only abst. (B5)	yes	0.967	0.971	0.953	0.958	1.548	0.452	0.063	0.000
	Random abst. (B5)	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	Random abst. (B5)	yes	0.967	0.971	0.953	0.958	1.548	0.452	0.063	0.000

Table 6. All methods at $\alpha = 0.2$, mean over predictors and seeds.

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
S1 (exchangeable)	Split (B1/B2)	no	0.786	0.797	0.675	0.787	0.927	0.919	0.282	0.000
	Split (B1/B2)	yes	0.782	0.796	0.680	0.779	0.923	0.921	0.287	0.000
	HCP (B3)	no	0.800	0.812	0.694	0.802	0.953	0.949	0.242	0.000
	HCP (B3)	yes	0.840	0.846	0.804	0.803	1.045	0.952	0.164	0.000
	Subsample once (B3)	no	0.800	0.812	0.692	0.803	0.954	0.947	0.243	0.000
	Subsample once (B3)	yes	0.840	0.846	0.802	0.804	1.046	0.951	0.165	0.000
	Repeated subsample (B3)	no	0.800	0.812	0.694	0.802	0.953	0.949	0.242	0.000
	Repeated subsample (B3)	yes	0.840	0.847	0.804	0.803	1.045	0.952	0.164	0.000
	GWCP (B4)	no	0.799	0.810	0.692	0.800	0.949	0.947	0.246	0.000
	GWCP (B4)	yes	0.839	0.845	0.803	0.801	1.043	0.953	0.166	0.000
	ProtCOVER (i)+(ii)	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.000
	ProtCOVER (i)+(ii)	yes	0.840	0.847	0.799	0.805	1.044	0.953	0.163	0.000
	ProtCOVER (i)+(iii)	no	0.800	0.812	0.694	0.802	0.953	0.949	0.242	0.000
	ProtCOVER (i)+(iii)	yes	0.840	0.847	0.804	0.803	1.046	0.951	0.164	0.001
	ProtCOVER full	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.001
	ProtCOVER full	yes	0.840	0.847	0.800	0.806	1.044	0.953	0.163	0.000
	Support-only abst. (B5)	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.001
	Support-only abst. (B5)	yes	0.840	0.847	0.799	0.806	1.044	0.953	0.163	0.000
	Random abst. (B5)	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.001
	Random abst. (B5)	yes	0.840	0.847	0.799	0.806	1.045	0.953	0.163	0.000
S2, $\beta = 1$	Split (B1/B2)	no	0.777	0.789	0.641	0.788	0.920	0.909	0.305	0.000
	Split (B1/B2)	yes	0.774	0.787	0.630	0.792	0.917	0.914	0.308	0.000
	HCP (B3)	no	0.790	0.803	0.658	0.801	0.942	0.941	0.266	0.000
	HCP (B3)	yes	0.829	0.836	0.754	0.811	1.024	0.967	0.187	0.000
	Subsample once (B3)	no	0.786	0.798	0.653	0.798	0.935	0.933	0.280	0.000
	Subsample once (B3)	yes	0.828	0.836	0.752	0.810	1.023	0.965	0.189	0.000
	Repeated subsample (B3)	no	0.790	0.803	0.659	0.801	0.943	0.941	0.266	0.000
	Repeated subsample (B3)	yes	0.828	0.836	0.754	0.810	1.023	0.967	0.189	0.000
	GWCP (B4)	no	0.794	0.806	0.663	0.804	0.948	0.943	0.259	0.000
	GWCP (B4)	yes	0.835	0.842	0.785	0.804	1.050	0.944	0.179	0.000
	ProtCOVER (i)+(ii)	no	0.800	0.813	0.672	0.810	0.960	0.950	0.246	0.000
	ProtCOVER (i)+(ii)	yes	0.844	0.850	0.800	0.810	1.075	0.921	0.162	0.000
	ProtCOVER (i)+(iii)	no	0.790	0.803	0.658	0.801	0.942	0.941	0.266	0.000
	ProtCOVER (i)+(iii)	yes	0.829	0.836	0.754	0.811	1.025	0.967	0.187	0.001
	ProtCOVER full	no	0.800	0.813	0.673	0.811	0.962	0.949	0.246	0.001
	ProtCOVER full	yes	0.844	0.850	0.801	0.811	1.075	0.921	0.162	0.000
	Support-only abst. (B5)	no	0.800	0.813	0.672	0.811	0.962	0.949	0.246	0.001
	Support-only abst. (B5)	yes	0.844	0.850	0.801	0.810	1.075	0.921	0.162	0.000
	Random abst. (B5)	no	0.800	0.813	0.672	0.811	0.962	0.949	0.246	0.001
	Random abst. (B5)	yes	0.844	0.850	0.801	0.810	1.075	0.921	0.162	0.000
S2, $\beta = 2$	Split (B1/B2)	no	0.776	0.788	0.633	0.789	0.920	0.907	0.306	0.000
	Split (B1/B2)	yes	0.773	0.786	0.618	0.794	0.917	0.913	0.312	0.000
	HCP (B3)	no	0.790	0.803	0.650	0.803	0.942	0.941	0.268	0.000
	HCP (B3)	yes	0.824	0.833	0.736	0.812	1.018	0.971	0.197	0.000
	Subsample once (B3)	no	0.788	0.801	0.648	0.801	0.939	0.937	0.274	0.000
	Subsample once (B3)	yes	0.825	0.834	0.736	0.813	1.020	0.967	0.196	0.000
	Repeated subsample (B3)	no	0.791	0.803	0.650	0.803	0.942	0.941	0.266	0.000
	Repeated subsample (B3)	yes	0.824	0.833	0.736	0.811	1.018	0.970	0.197	0.000
	GWCP (B4)	no	0.798	0.810	0.661	0.810	0.957	0.941	0.254	0.000
	GWCP (B4)	yes	0.830	0.837	0.778	0.798	1.054	0.931	0.192	0.000
	ProtCOVER (i)+(ii)	no	0.808	0.821	0.675	0.819	0.979	0.945	0.229	0.000
	ProtCOVER (i)+(ii)	yes	0.842	0.850	0.796	0.810	1.079	0.913	0.165	0.000
	ProtCOVER (i)+(iii)	no	0.790	0.803	0.650	0.803	0.942	0.941	0.268	0.000
	ProtCOVER (i)+(iii)	yes	0.825	0.833	0.736	0.812	1.019	0.970	0.196	0.001
	ProtCOVER full	no	0.811	0.824	0.684	0.821	0.998	0.928	0.225	0.018
	ProtCOVER full	yes	0.845	0.853	0.800	0.813	1.096	0.897	0.162	0.019
	Support-only abst. (B5)	no	0.811	0.824	0.682	0.822	0.998	0.928	0.225	0.018
	Support-only abst. (B5)	yes	0.845	0.852	0.799	0.813	1.094	0.899	0.162	0.019
	Random abst. (B5)	no	0.811	0.824	0.683	0.822	0.999	0.927	0.225	0.018
	Random abst. (B5)	yes	0.845	0.852	0.799	0.813	1.095	0.898	0.162	0.019
S3 (orphan families)	Split (B1/B2)	no	0.778	0.789	0.664	0.781	0.924	0.912	0.300	0.000
	Split (B1/B2)	yes	0.774	0.788	0.664	0.775	0.919	0.916	0.307	0.000
	HCP (B3)	no	0.794	0.805	0.685	0.797	0.951	0.946	0.257	0.000
	HCP (B3)	yes	0.836	0.846	0.799	0.796	1.040	0.955	0.174	0.000
	Subsample once (B3)	no	0.795	0.806	0.682	0.801	0.954	0.943	0.255	0.000
	Subsample once (B3)	yes	0.835	0.846	0.799	0.794	1.039	0.958	0.176	0.000
	Repeated subsample (B3)	no	0.794	0.806	0.686	0.797	0.951	0.946	0.257	0.000
	Repeated subsample (B3)	yes	0.836	0.846	0.798	0.796	1.040	0.956	0.175	0.000
	GWCP (B4)	no	0.802	0.814	0.695	0.806	0.967	0.957	0.236	0.000
	GWCP (B4)	yes	0.839	0.847	0.792	0.805	1.046	0.948	0.170	0.000
	ProtCOVER (i)+(ii)	no	0.807	0.819	0.700	0.810	0.976	0.956	0.227	0.000
	ProtCOVER (i)+(ii)	yes	0.843	0.850	0.793	0.812	1.056	0.939	0.163	0.000
	ProtCOVER (i)+(iii)	no	0.794	0.805	0.685	0.797	0.951	0.946	0.257	0.000
	ProtCOVER (i)+(iii)	yes	0.837	0.847	0.800	0.797	1.044	0.951	0.173	0.005
	ProtCOVER full	no	0.807	0.819	0.701	0.810	0.977	0.955	0.227	0.001
	ProtCOVER full	yes	0.843	0.850	0.793	0.812	1.058	0.937	0.162	0.003
	Support-only abst. (B5)	no	0.807	0.819	0.701	0.810	0.977	0.955	0.227	0.001
	Support-only abst. (B5)	yes	0.843	0.850	0.793	0.812	1.058	0.937	0.162	0.003
	Random abst. (B5)	no	0.807	0.819	0.701	0.810	0.977	0.955	0.227	0.001
	Random abst. (B5)	yes	0.843	0.850	0.793	0.812	1.058	0.937	0.162	0.003